

Figure 5 Gate

Drug Exposure, the Hydrolysate Inhibitor Figure, as Mockups

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Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Figure and table status: ● on file, every value in a committed result file ▮ partial, a run unfinished or unscored ○ idea only, a sketch or nothing run

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1 The decision ✗

Manuscript section R5 covers drug exposure: a compound and its dose as a perturbation the model reads alongside the genotype. The case study is a hydrolysate. Sorghum hydrolysate carries six inhibitors that a fermentation host meets together, and the private Bioscreen runs of 2021 measured all 6 of them on one strain, bAID in aerobic YPD: 180 titration wells, 197 wells over 63 combinations, and 600 wells over 3 isobole grids, 977 wells in all. The models that would predict them were trained on the pooled chemogenomic table: 4 public screens, 6,042,771 cells, 5,463 compounds over 5,863 genes. Nothing here is approved. The sheets are mockups of what the result files support, and the symbol after each panel letter says only whether the values behind it are on file. Color is the source of a predicted value: amber, a measured value, whether a public screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel's legend names it. A label with no bar is an arm that has not run.

What the figure would claim. A model trained on public chemogenomic screens can predict how an industrial host behaves under a mixture of inhibitors it has never seen, and it can do so well enough to rank the combinations a process should avoid. The figure is built to make that claim testable rather than to assert it: the bottom row puts the model beside the two model-free references on the same wells, so a reader sees what the model adds over arithmetic on the single-agent curves, including when the answer is nothing.

The four claims, and where each one stands. First, that growth under a combination departs from independence. On file: over the 63 combinations the Loewe rule reaches a growth AUROC of 0.958 and an accuracy of 0.825, Bliss from the observed singles 0.872 and 0.556, and the highest single agent 0.663; over the 21 combinations that grew, Bliss ranks fitness at Spearman 0.573 with a mean signed error of -0.142 and Loewe at 0.713 with 0.128. Of 15 pairs, 12 are synergistic relative to Bliss, 2 additive and 1 antagonistic. Second, that chemogenomic profile similarity anticipates which pairs deviate. This FAILS as pre-registered: with measured profiles in the matrix the association is 0.204 at a permutation p of 0.462 over 15 pairs, and the all-predicted matrix carries the opposite sign, -0.686 at p 0.006 on raw profiles, which the fingerprint explains rather than the biology. Third, that deletion profiles compose under a mixture by a rule measurable on public data. On file: over 26 Hillenmeyer HET pairs the mean of the singles is the best fixed rule in 24 of them, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single, and the free linear fit puts weight 0.529 on one single and 0.593 on the other. Fourth, post-analysis only: composed profiles nominate the genes a screen under each combination would recover, 68 core genes from the best-available matrix and 105 from the all-predicted one. No screen exists to test them.

The bars a model has to clear. Growth: AUROC 0.958, the Loewe rule from the ex21 Hill fits. Fitness over the combinations that grew: Spearman 0.573, Bliss from the observed ex23 singles. Both are model-free, both come from the same wells the model would be scored on, and both are already in a committed result file. On the public side the bar is the nested ridge on fingerprint counts at a median centered Spearman of 0.359 over 96 compound-cold folds, which the next best arm of the ladder trails by a hundredth (0.357).

Where the full record is. The sheets draw the figure; the measurements behind them are written up elsewhere, and the source table under each figure points to the file. `notes-tex/wet-lab/040-inhibitor-synergy-wetlab` is the case-study document: the 2021 runs, the growth calls, the four claims scored, and the confounds each caption carries. `notes-tex/metabolism/031-betaxanthin-module` and `notes-tex/metabolism/032-amino-acid-module` hold the campaign whose panel conventions this document copies. The chemogenomic side is in three experiment records rather than documents: `experiments/033-env-chemgen-pooled` is the pooled table and its axes, `experiments/035-env-chemgen-vanacloig-cgt` the first compound-cold campaign and the learning curve, and `experiments/038-env-chemgen-vanacloig-cgt-corrected` the ladder and the factorized rounds on the corrected store. The 035 and 038 branches are not landed, so the panel script reads their result files by absolute path and records every path it read in `experiments/040-inhibitor-synergy-wetlab/results/figure_gate_manifest.json`.

What a decision here settles. Whether the figure is the four-by-three of fig. 2 or the three-by-three the experiment note proposes, which of the twelve panels earns main-text space against supplementary space, and whether the bottom row ships with the model beside the rules or with the rules alone. Nothing in this document is a publication-ready panel; the panels are drawn at Nature print width so that the argument, not the artwork, is what gets judged.

2 The mockups ✗

Each caption gives, per panel, what is plotted, the statistic and the status; the table under each figure gives the files behind it. The statistic on the public side is always the per-compound centered Spearman under compound-cold folds, median over compounds, with 26 compounds fitted per fold; on the wet-lab side it is the growth AUROC over the 63 combinations and the fitness Spearman over the 21 that grew under the served call. Color is the source of a predicted value: amber, a measured value, whether a public screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel’s legend names it. A label with no bar is an arm that has not run.

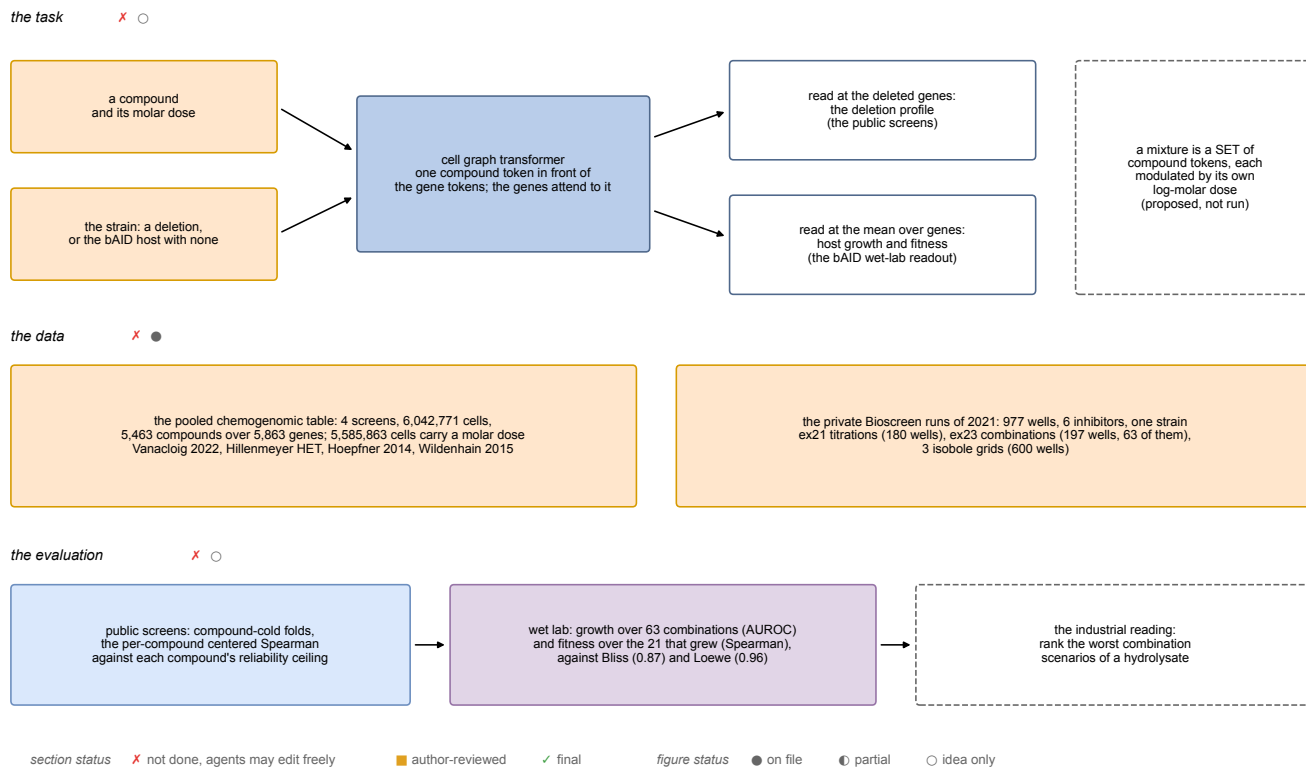


Figure 1: The task, the data and the evaluation. Three blocks, each a layout rather than a drawn result; a dashed box is proposed and not run. **The task** ✗ ○: a compound and its molar dose enter the cell graph transformer as one token placed in front of the gene tokens, which attend to it; a public-screen strain is read at its deleted genes, giving a deletion profile, and the bAID host, which carries no deletion, is read at the mean over genes, giving growth and fitness. A mixture becomes a set of compound tokens, one per inhibitor, each modulated by its own log-molar dose, and that is the part nothing has yet run. **The data** ✗ ●: the pooled chemogenomic table over 4 screens, 6,042,771 cells and 5,463 compounds across 5,863 genes, of which 5,585,863 cells carry a molar dose, beside the private Bioscreen runs, 977 wells over 6 inhibitors on one strain. **The evaluation** ✗ ○: compound-cold folds and the per-compound centered Spearman on the public side, growth and fitness against Bliss and Loewe on the wet-lab side, and the industrial reading, which is to rank the worst combination scenarios of a hydrolysate.

Sources for fig. 1

the task	Nothing drawn from a file. The encoder it describes is the environment-encoder head of 038/; the host head is the proposed adaptation of <code>experiments/040-inhibitor-synergy-wetlab/scripts/train_mixture.py</code> . Status: idea.
the data	033/pool_totals.csv, 033/store_axes.csv (the screens, the cells, the compounds, the genes, the molar-dose coverage); wetlab_wells.csv and wetlab_provenance.json (the wells, the runs, the raw-mirror shas). Status: on file.
the evaluation	038/ladder/ladder_r2_scores.csv (the fold size); mixture_scores.csv (the two bars). The boxes themselves are a layout, so the block is marked idea even though its counts are on file.

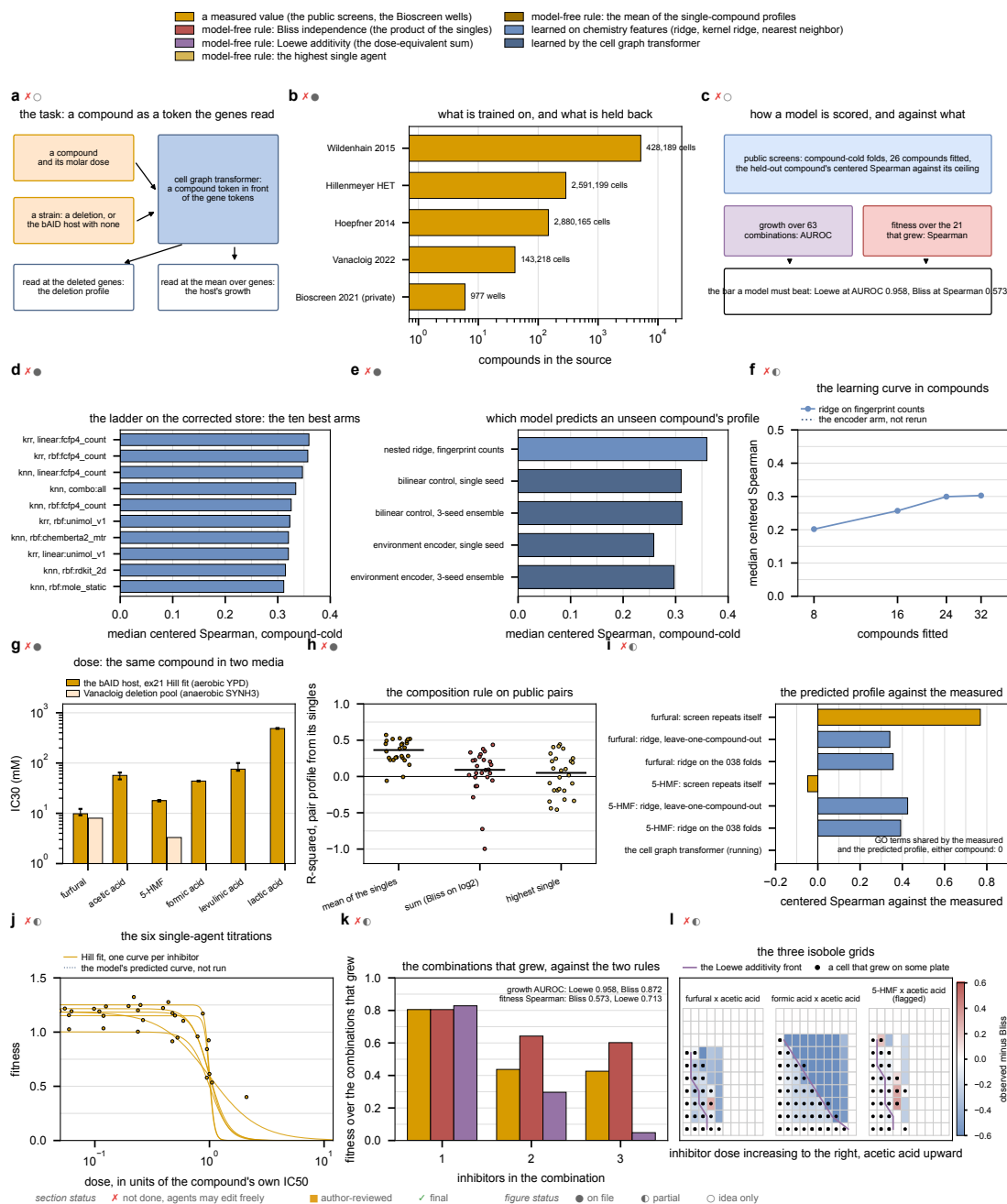


Figure 2: The sheet proposed for the main text, the four-by-three of the experiment note. **a**, The task as one strip; a draw.io composition replaces it. $\times \circ$ **b**, What is trained on: compounds per source on a logarithmic axis, each bar annotated with the cells or wells behind it. The Bioscreen runs carry 6 compounds against 5,463 in the pooled table, and the screen supplying the deletion profiles carries 41. $\times \bullet$ **c**, How a model is scored, and the two bars it must beat; a layout whose numbers are on file. $\times \circ$ **d**, The ladder, ten best arms of 67: nested ridge on fingerprint counts reads 0.359, the next arm 0.357, the tenth 0.311. $\times \bullet$ **e**, Which model predicts an unseen compound's profile. The second round is complete: over 96 compound-evaluations, 32 compounds on each of three fold seeds, the three-seed ensemble of the bilinear control reads 0.312 and of the environment encoder 0.297, both below nested ridge at 0.359. Paired on compound and fold, the encoder minus its control is -0.036 with 37 of 96 up. $\times \bullet$ **f**, The learning curve in fitted compounds, 0.202 at eight to 0.303 on the pool, about 0.049 per doubling. $\times \bullet$ **g**, Dose, the host's potency against the screen's: furfural 9.82 millimolar against 8, ratio 1.23; 5-HMF 17.9 against 3.30, ratio 5.42; the other four are in no screen. $\times \bullet$ **h**, How a pair's profile composes from its singles over 26 public pairs: the mean is best in 24, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single. $\times \bullet$ **i**, The predicted profile against the measured one. Furfural: the screen repeats itself at 0.769, ridge reads 0.341; 5-HMF: -0.048, which is noise, against 0.425. The two profiles share 0 significant terms. $\times \bullet$ **j**, The six titrations with their Hill fits, dose in units of each compound's own half-maximal dose; slopes run 1.21 to 20.0. $\times \bullet$ **k**, The combinations that grew, observed against the two rules, by number of inhibitors. Growth AUROC: Loewe 0.958, Bliss 0.872. Fitness Spearman: Bliss 0.573, Loewe 0.713. $\times \bullet$ **l**, The 3 isobole grids as observed minus Bliss, the Loewe front drawn and the cells that grew marked. Mean excess: -0.082, -0.446 and -0.017, the last a single run with growth islands and flagged. $\times \bullet$ **Partial**: no model front.

Sources for fig. 2

a	Nothing from a file; a layout of the proposed model. Status: idea.
b	033/store_axes.csv, 033/pool_totals.csv, wetlab_wells.csv. Status: on file.
c	038/ladder/ladder_r2_scores.csv, mixture_scores.csv; the boxes are a layout. Status: idea.
d	038/ladder/ladder_r2_summary.csv, centered target, full compound count. Status: on file.
e	038/ladder/ladder_r2_summary.csv; 038/factorized/r9_control/*_scores.csv; 038/factorized/r10_envenc/*_scores.csv. Two folds per arm were on disk when the sheet was drawn; the round was still running. Status: partial.
f	035/learning_curve/ridge_curve.csv, centered target. The encoder arm of the curve has no file. Status: partial.
g	single_agent_fits.csv (the host, with bootstrap intervals from single_agent_bootstrap.csv); vanacloig_ic30.csv (the screen). Status: on file.
h	het_rule_fit.csv, het_summary.json. Status: on file.
i	profile_predicted_vs_measured.csv (loco_32, centered); go_overlap_measured_vs_predicted.csv. The transformer column has no file. Status: partial.
j	wetlab_wells.csv, single_agent_fits.csv, single_agent_bootstrap.csv. No model curve exists. Status: partial.
k	mixture_combinations.csv, mixture_scores.csv, served call. No model arm exists. Status: partial.
l	isobole_cells.csv, isobole_summary.csv. No model front exists. Status: partial.

learned on chemistry features (ridge, kernel ridge, nearest neighbor)

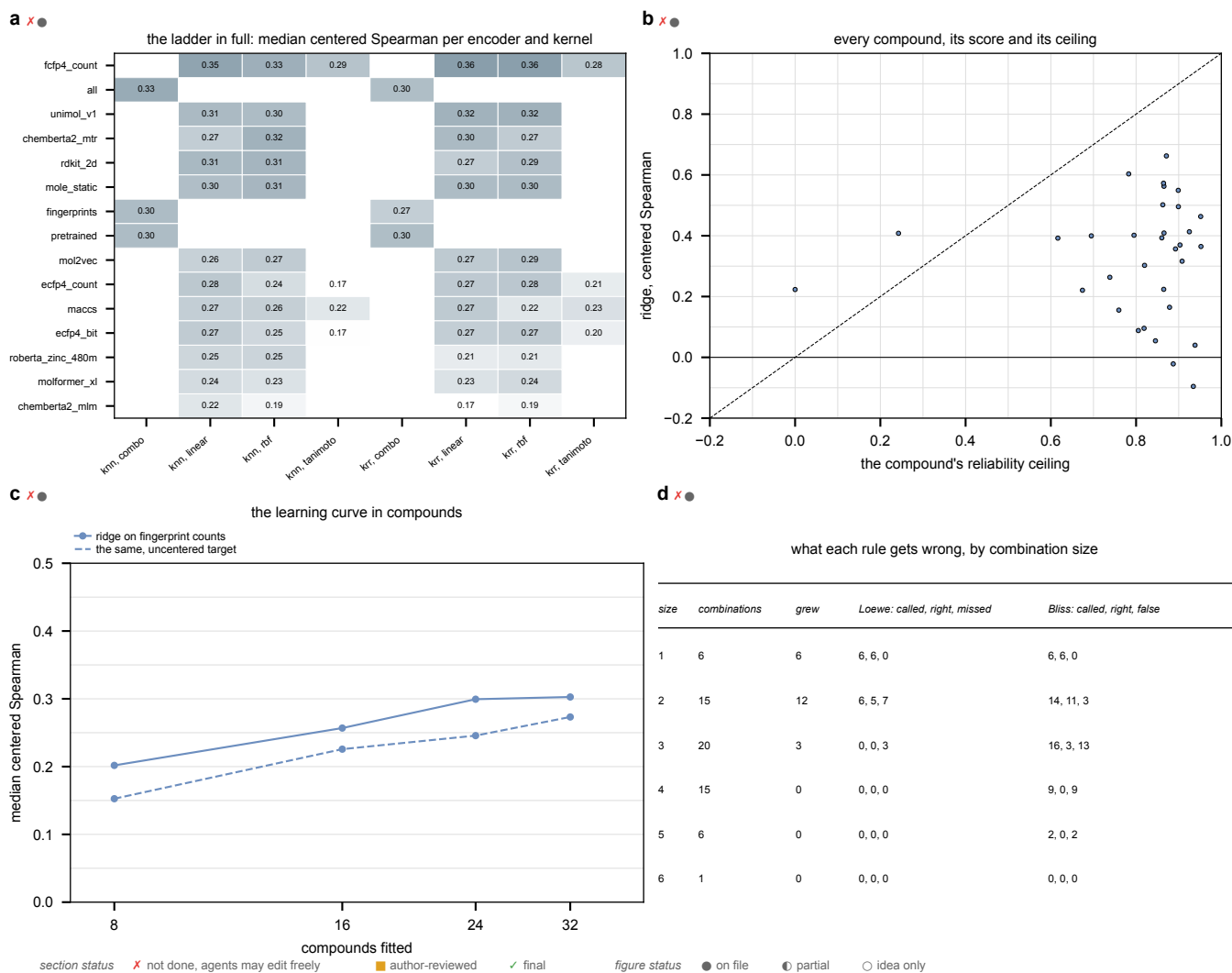


Figure 3: Supplement one: the public benchmark in full. **a**, Every ladder arm, median centered Spearman per encoder and kernel, 62 fitted cells over 15 encoders. Fingerprint counts lead every kernel they are defined for. **b**, Each of the 32 compounds, its ridge score against its own reliability ceiling: median 0.367, 2 above the ceiling and 2 below zero. **c**, The learning curve on both targets, the uncentered one dashed. The centered curve is the one the main sheet carries. **d**, What each rule gets wrong, by combination size: how many combinations the rule called alive, how many of those were right, and what it missed or invented. Loewe misses all 3 growing triples and 7 of 12 growing pairs; Bliss calls 13 of 17 dead triples alive.

Sources for fig. 3

- a 038/ladder/ladder_r2_summary.csv. Status: on file.
 - b 038/ladder/ladder_r2_scores.csv, kernel ridge on fingerprint counts, centered, averaged over fold seeds. Status: on file.
 - c 035/learning_curve/ridge_curve.csv. Status: on file.
 - d mixture_scores.csv, served call, one row per combination size. Status: on file.
-

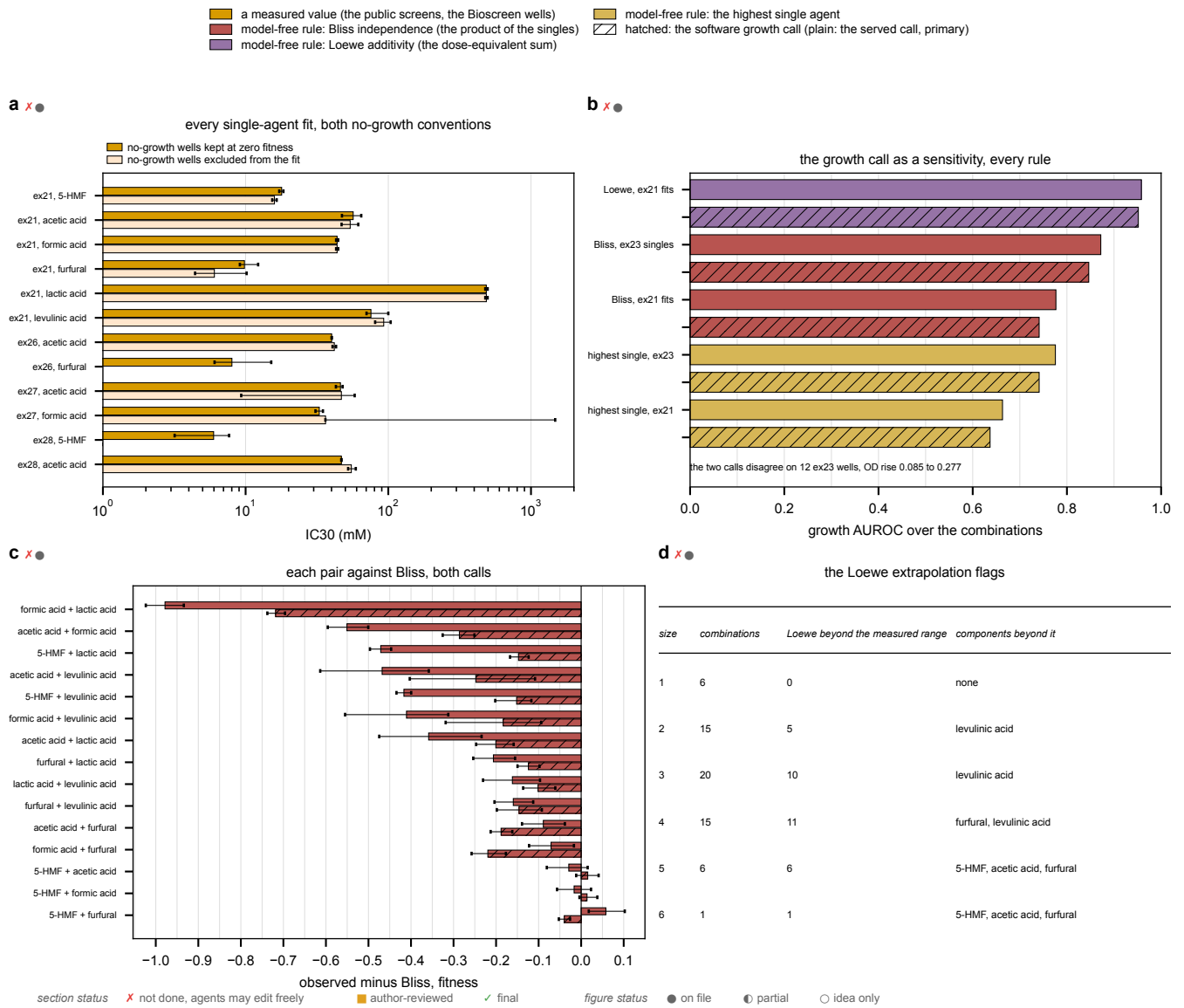


Figure 4: Supplement two: the single agents and the growth call. **a**, Every single-agent fit, 24 of them over the titration run and the three isobole axes, under both no-growth conventions: the darker bar keeps no-growth wells at zero fitness, the lighter one drops them from the fit. Intervals are the bootstrap. **b**, Every claim-one score under both growth calls, hatched for the software call. The two calls disagree on 12 wells of the combination run across 4 combinations, with optical-density rises from 0.085 to 0.277. **c**, Each pair against Bliss with its replicate-bootstrap interval, under both calls. Under the served call 12 of 15 pairs are synergistic, 2 additive and 1 antagonistic. **d**, Where the Loewe rule had to extrapolate past the measured dose range: 33 of the combinations, and 0 of the isobole fronts fell outside their grid.

Sources for fig. 4

- a `single_agent_fits.csv`, `single_agent_bootstrap.csv`. Status: on file.
 - b `mixture_scores.csv` (both calls), `wetlab_growth_call_check.csv`. Status: on file.
 - c `pair_deviation.csv`. Status: on file.
 - d `mixture_combinations.csv` (the extrapolation flags), `isobole_summary.csv`. Status: on file.
-

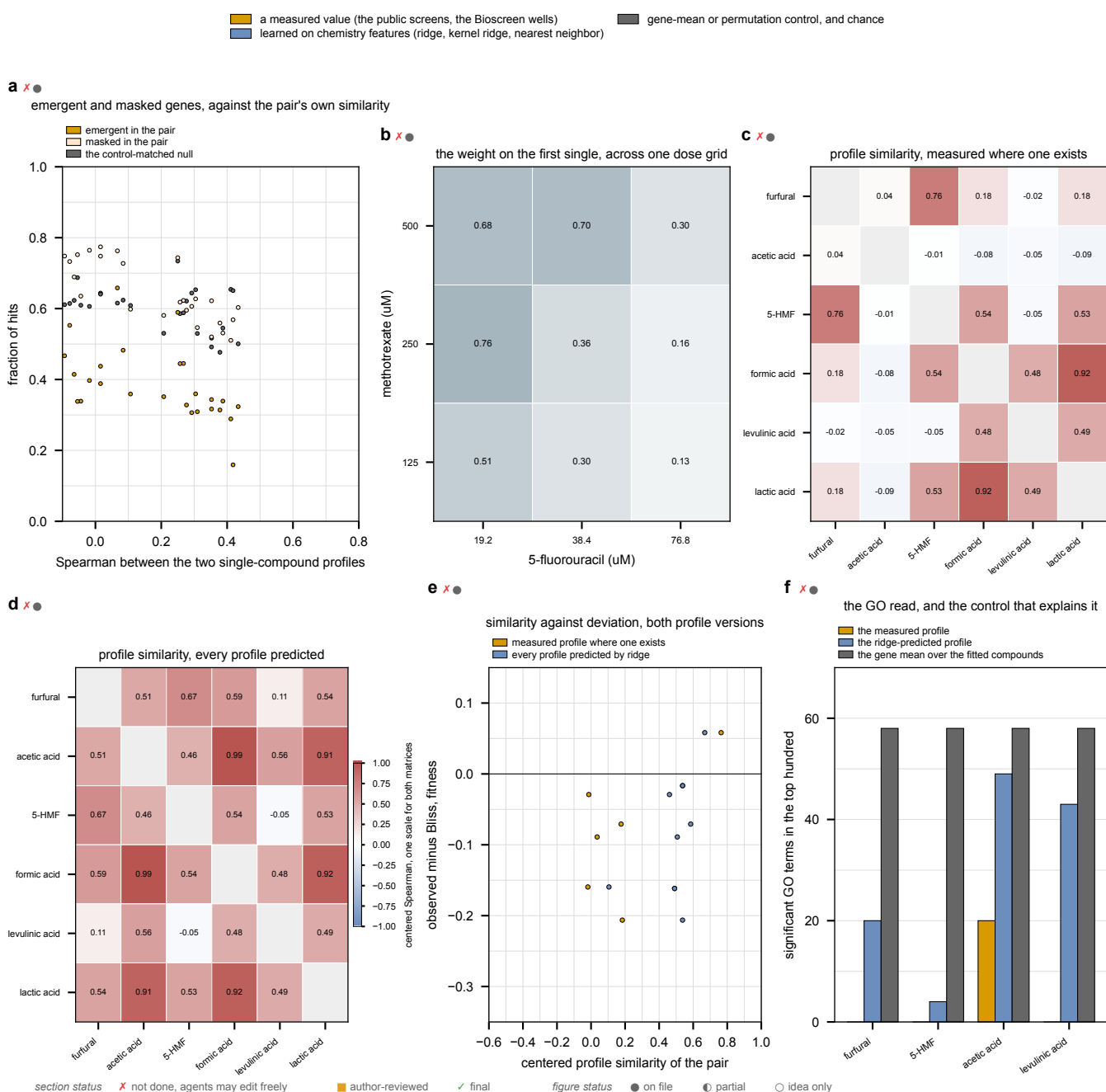


Figure 5: Supplement three: how profiles compose, and what the predicted ones carry. **a**, Emergent and masked gene rates of each public pair against how alike its two singles are, with the control-matched null beside them. The median emergent fraction is 0.355 against a null of 0.621, and the masked fraction 0.649; the only two near-replicate singles in the screen disagree on 0.536 to 0.603 of their own hits, which is the noise floor these rates sit inside. ✕ ● **b**, The one dose grid the public screen holds, methotrexate against fluorouracil: the weight the free linear fit puts on the first single moves from 0.128 to 0.757 across 9 dose pairs, while the mean wins at 9 of them. ✕ ● **c**, **d**, Profile similarity among the six inhibitors, measured where a measurement exists and then with every profile predicted. Measured, the strongest pair reads 0.924; predicted, 0.986, because the small acids have near-identical predicted profiles. One scale carries both panels. ✕ ● **e**, Similarity against the pair's departure from Bliss, both versions: 0.204 at a permutation p of 0.462 with measurements in, and -0.225 at 0.425 with everything predicted. ✕ ● **f**, The GO read: significant terms in the top hundred sensitive genes of the measured profile, of the ridge-predicted profile, and of the gene mean over the fitted compounds. The control carries 58 terms and the predicted profiles share 48 of them, which is what the predicted enrichment is. ✕ ●

Sources for fig. 5

a	het_emergent_masked.csv, het_near_replicates.csv, het_summary.json. Status: on file.
b	het_rule_fit.csv, the methotrexate and fluorouracil family. Status: on file.
c, d	profile_similarity.csv, centered target, both versions. Status: on file.
e	similarity_vs_deviation_pairs.csv, similarity_vs_deviation.csv. Status: on file.
f	go_overlap_measured_vs_predicted.csv, nominations_summary.csv. Status: on file.

3 What each panel still needs ✗

Table 1: What each panel still needs before it is real. The status column repeats the panel’s mark in words: on file (filled circle), every drawn value is in a committed result file; partial (half circle), a run is unfinished or a planned arm has no file; idea only (open circle), a layout or nothing run. No panel is approved; approval is the section status and is the author’s.

panel		what remains
concept; main, a	○	Redraw as one draw.io composition at print size. The counts in it are on file; the boxes and arrows are a layout, and the mixture path they describe (a set of compound tokens, each modulated by its own log-molar dose) is the proposed model and has not run.
main, c	○	Same: a layout whose numbers are on file. It merges into the concept sheet if the figure is cut from four rows to three.
main, e; SI-1, a	◐	The second factorized round is producing. Two folds of the bilinear control and two of the environment encoder were on disk when the sheet was drawn; the round’s remaining folds, the nine-seed ensembles and the stack with nested ridge are not written yet. Round eleven on Delta adds the width and depth axis.
main, f	◐	The encoder arm of the learning curve has to be rerun on the corrected store; only the ridge arm has a file. The planned round also adds the other pooled sources to the gene-level task, which is the second line this panel would carry.
main, i	◐	The transformer’s compound-cold profiles for furfural and 5-HMF come out of the same running round. Until then the panel shows ridge against the measured profile and the screen’s own repeat, and the transformer row is a label with no bar.
main, j	◐	The model’s predicted single-agent curves do not exist. They need the host head fitted on the titration run, in both variants the plan names: trained on the public dose anchors alone, and calibrated on the titration run.
main, k, l	◐	The model’s predictions over the combinations and over the isobole grids do not exist. The rules are on file and are the bars. If the model does not beat them, the panels ship with the rules and the model side by side and the text says so.
main, b, d, g, h	●	Nothing. Every drawn value is in a committed result file. These four are what the figure can claim today.
SI-1, b, c, d	●	Nothing.
SI-2, all	●	Nothing.
SI-3, all	●	Nothing, with one caveat the caption carries: the similarity matrices mix a measured profile for two inhibitors with predicted profiles for four, and the all-predicted version exists so that mixture is visible rather than hidden.

The running and planned rounds, and what each one fills. The second factorized round of the corrected-store campaign is on the two GilaHyper cards named for it, one job for the bilinear control and one for the environment encoder; it fills main **e**, main **i** and the first column of SI-1 **a**. Round eleven of the same campaign goes to Delta under the account the plan names, never a preemptible partition, and widens the ladder rather than deepening it. The first mixture round of this experiment is the one that fills the bottom row: about ten arms by three seeds, crossing the pooled sources against the host-training recipe (the public dose anchors alone, the anchors plus the titration run, and a fine-tune on the titration run) against the dose treatment. It starts once the mixture trainer passes its processor smoke test, and its launchers are the pair of scripts the experiment folder already carries.

The cut from four rows to three. The experiment note proposes merging rows two and three. The merge that costs least is to move main **c** into the concept sheet, where the same boxes already appear, and to put main **f** and main **g** side by side as one panel on two axes, since both are about what more data buys: more compounds on one axis, a second medium on the other. That leaves nine panels: the task, the data, the ladder, the transformer against ridge, the merged data-and-dose panel, the composition rule, the predicted profile, the single agents, and the combinations with the isoboles as its inset. The alternative, dropping the single-agent panel, is worse: it is what every rule in the bottom row is computed from, and a reader who cannot see the Hill slopes cannot judge why Bliss and Loewe disagree as far as they do.

4 State on the day the sheets were drawn x

What had finished. The processor round of this experiment: the four pre-registered claims scored, which is every panel marked on file above. The ladder of the corrected-store campaign, giving main **d** and SI-1 **a**. The learning curve of the earlier campaign, giving main **f** in its ridge arm. The pooled table and its axes, giving main **b**.

What finished while the sheets were drawn. The second factorized round of the corrected-store campaign, submitted in the small hours of the same day on two GilaHyper cards: one job for the bilinear control, one for the environment encoder, each fifteen configurations of five compound-cold folds on three fold seeds, three seeds per fold. Both completed, giving main **e** in full: the control's three-seed ensemble at 0.312 and the encoder's at 0.297 over 96 compound-evaluations, against nested ridge at 0.359, with the encoder -0.036 against its own control when the two are paired on compound and fold. Neither head reaches the fingerprint ridge on this store, which reverses the earlier campaign's reading and is the result the two Delta rounds below are meant to explain or confirm. The sheet re-reads the directory every time it is drawn, so a later round replaces these values rather than adding to them.

What had not started. The first mixture round, which is every panel of the bottom row in its model arm. The trainer and its configuration groups are owned by the implementation thread and are not part of this document.

What to expect, labeled. Hypothesis (untested): the environment encoder ties nested ridge on the corrected store, as it did on the earlier store, so main **e** closes with the two within a hundredth of each other and the figure's public-benchmark row says that architecture is not where the gain is. Hypothesis (untested): the encoder's compound-cold profiles for the two inhibitors in store read no better than ridge, so main **i** keeps its shape and the predicted profiles stay generic. Hypothesis (untested): a dose-aware host head fitted on the titration run beats the highest single agent on growth, which reads 0.663, and does not beat Loewe, which reads 0.958, because Loewe already uses the fitted single-agent curves and the model has no information about this host that those curves do not carry. If that last one holds the honest figure is the model-free one, and the bottom row says so in the caption rather than in a footnote.

What would change the figure's claim. A model arm that beats 0.958 on growth, or 0.573 on fitness over the combinations that grew, moves the bottom row from a validation of arithmetic to a result about transfer. A model arm that does neither still earns the row, because the comparison is the point: the rules are computed from six titrations on the same strain, and a model trained on 5,463 public compounds that cannot beat them is the measurement the section needs to report.